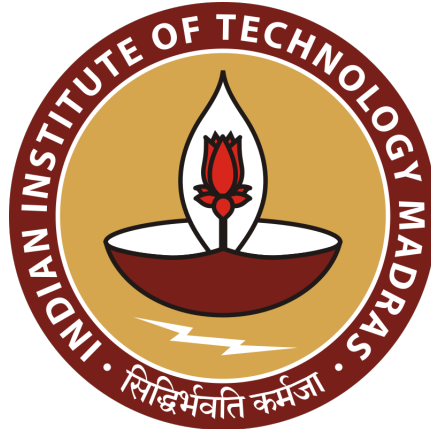


INDIAN INSTITUTE OF TECHNOLOGY, MADRAS



Data Science and AI Project

MILESTONE 6 SUBMISSION

NON-TECHNICAL REPORT

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1 Project Overview (Non-Technical Summary)

1.1 Introduction

Forests play a crucial role in maintaining ecological balance, supporting biodiversity, and regulating climate. However, in many regions of India, especially in the North-Eastern states, forests are continuously affected by activities such as shifting cultivation and land clearing. Monitoring these changes manually is slow, expensive, and often inaccurate.

This project aims to develop an automated system that can detect changes in forest cover using satellite images. By comparing images taken at two different time periods, the system identifies areas where forest loss has occurred.

1.2 Objective

The primary goal of this project is to:

- Detect forest loss automatically using satellite data
- Reduce the need for manual inspection
- Provide faster and more consistent monitoring
- Support decision-making for environmental protection

1.3 How the System Works

The system takes satellite images of the same location from two different years (for example, 2021 and 2023). It then compares these images to find differences.

In simple terms, the model:

- Looks at “before” and “after” images
- Identifies changes in vegetation and land patterns
- Highlights areas where forests may have been cleared

The output is a map that shows:

- Areas with no change
- Areas where forest loss has likely occurred

1.4 Key Findings

The system was tested on a large dataset of satellite images. The results show that:

- The model can successfully identify major areas of forest loss
- It performs better when the changes are large and clearly visible
- Detecting very small or scattered changes remains challenging

The final version of the model achieved reliable performance and produced results that closely match real-world observations.

1.5 Impact and Applications

This project has several important real-world applications:

- **Environmental Monitoring:** Helps track deforestation and land-use changes over time
- **Government Support:** Assists policymakers in planning conservation strategies
- **Disaster Management:** Useful for identifying areas affected by natural or human-induced changes
- **Sustainable Development:** Supports efforts to balance development with environmental protection

1.6 Limitations

While the system performs well in many cases, it has some limitations:

- It may miss very small or subtle changes in forest cover
- Some natural features like bare soil or shadows can be confused with forest loss
- The accuracy depends on the quality and quantity of available data

1.7 Future Scope

There is significant potential to improve and expand this system:

- Use more data from different regions and time periods
- Improve detection of small and gradual changes
- Incorporate additional information such as weather or seasonal patterns

- Develop real-time monitoring tools for faster decision-making

1.8 Conclusion

This project demonstrates how artificial intelligence can be used to monitor environmental changes efficiently. By automating the detection of forest loss, the system can save time, reduce human effort, and provide valuable insights for protecting natural resources.

With further improvements and larger datasets, such systems can play a key role in addressing environmental challenges and supporting sustainable development.